

Introduction and Different Flavors of Selenium

Course: Selenium Web Automation Training

Module / Week: Week 1 Assignment

1. What, Why Selenium

What is Selenium?

Selenium is a open-source automated testing framework used to validate web applications across different browsers in different operating systems using different programming languages. It allows developers and software testers to write automated scripts in multiple programming languages to interact with different web browsers.

Why Use Selenium?

- I. **Open-Source & Free:** Selenium has no licensing fees or hidden costs. It has a large global open source community.
- II. **Multi-Language Support:** Selenium scripts can be written in Python, Java, C#, JavaScript, Ruby, and Kotlin.
- III. **Multi-Browser Support:** Selenium is Compatible with Chrome, Firefox, Microsoft Edge, Safari and Opera.
- IV. **Cross-Platform Support:** Selenium can run seamlessly on different operating systems like Windows, macOS and Linux.
- V. **Ecosystem & Framework Integration:** Selenium can be integrated smoothly with testing frameworks (pytest, unittest, TestNG, JUnit), CI/CD tools (Jenkins, GitHub Actions) and Docker.

2. Selenium IDE and Selenium RC

Selenium IDE (Integrated Development Environment)

Selenium IDE is a browser extension (for Chrome, Firefox, and Edge) that provides a simple record-and-playback interface for web automation. It requires no advanced programming knowledge to get started.

Key Use Cases: Quick test prototyping, reproducing bugs quickly and exporting test suites into programming languages.

Limitations: Lacks support for complex conditional logic, iterations, heavy data-driven testing, and parallel execution.

Selenium RC (Remote Control / Selenium 1)

Selenium RC was the original flagship component of Selenium before Web Driver was created. It used a JavaScript injection server (Selenium RC Server) that acted as an HTTP proxy between test commands and the browser.

Architecture: Injected a JavaScript program (Selenium Core) into the browser to execute user actions.

Limitations: Slow performance, complex setup, and subject to browser Same-Origin Policy (SOP) restrictions.

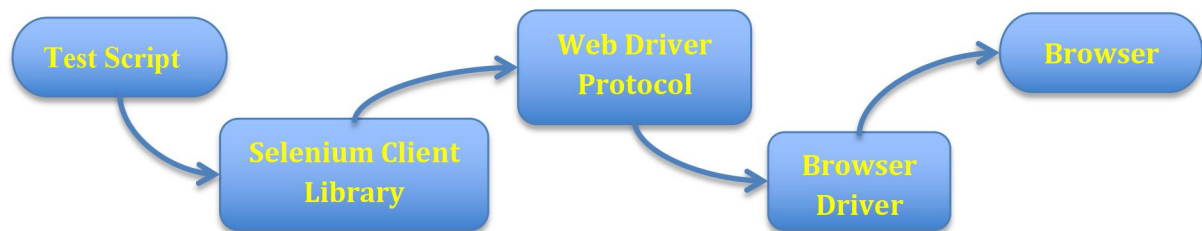
Current Status: Deprecated and obsolete; merged with Web Driver in Selenium 2.0.

3. Selenium WebDriver

Selenium Web Driver (Selenium 2, 3, and 4) is the core component of modern Selenium. Unlike Selenium RC, Web Driver communicates directly with the browser engine driver without injecting JavaScript into the page.

Web Driver Architecture & Workflow

The basic Selenium Web Driver architecture consists of these components:



- I. **Test Script** : It is written using a programming language such as Python, Java, C# or JavaScript. It contains commands such as `click()`, `send_keys()`, and `find_element()`.
- II. **Selenium Client Library**: Converts the programming language commands into Web Driver requests.
- III. **W3C Web Driver Protocol**:
 - a. **JSON Wire Protocol**: Older, Selenium specific protocol used for communication between Web Driver and browsers.
 - b. **W3C Web Driver Protocol**: Modern standardized protocol that provides more consistent and cross browser compatible communication.
- IV. **Browser Driver**: Translates commands into native browser commands. Acts as a bridge between Selenium and the browser. ChromeDriver (Chrome), Gekodriver (Firefox), EdgeDriver (Microsoft Edge) are some widely used Browser drivers.
- V. **Browser**: It receives instructions from the driver. Performs actions such as opening pages, clicking buttons, entering text and others. Chrome, Firefox, Edge these are some widely used web browsers.

Key Strengths of Web Driver

- I. **Advanced Interactivity:** Handles dynamic elements, frames, alerts, windows/tabs, mouse drag-and-drop, and implicit/explicit waits.
- II. **Direct Browser Control:** Controls browsers directly for faster and accurate automation.
- III. **Cross-Browser Support:** Works with Chrome, Firefox, Edge, Safari, and more.
- IV. **Multi-Language Support:** Supports Python, Java, C#, JavaScript, Ruby, and Kotlin.
- V. **Advanced User Interactions:** Handles alerts, frames, windows, mouse actions, and keyboard events.
- VI. **W3C Standard:** Provides standardized and consistent browser automation.

4. Selenium GRID

Selenium Grid is a tool that allows automated tests to run simultaneously across multiple browsers, operating systems, and machines.

Why Use Selenium Grid?

- I. **Parallel Execution:** Run multiple tests at the same time.
- II. **Cross-Browser Testing:** Test on Chrome, Firefox, Edge, etc.
- III. **Multiple Platforms:** Test across different operating systems.
- IV. **Faster Execution:** Reduces overall testing time.